IT Service Desk Support & Windows Troubleshooting Lab

Level: Beginner / L1 | Tools: Windows, Command Prompt, Excel | Duration: 3–5 days

How to use this document: Run each scenario on your own Windows PC or a virtual machine, then replace every field in [brackets] with what you actually observed. Only claim on your resume what you really performed. This is a lab project, not professional experience.

Part 1: Troubleshooting Log

Copy one log entry per scenario run. Fill in the real results.

Log 1: Slow computer / application not responding (Windows)

- **Date / time:** [date]
- **User report (simulated):** "My laptop is very slow and apps keep freezing."
- **Environment:** [Windows 11 / VM name / RAM / disk type]
- **Checks performed:**
  1. Opened Task Manager (Ctrl + Shift + Esc).
  2. Recorded CPU: [x]%, Memory: [x]%, Disk: [x]%.
  3. Identified top resource consumers: [process names].
  4. Checked Settings > Windows Update for pending updates: [yes/no].
  5. Saved work, then restarted the affected application: [app name].
- **Observations:** [e.g., browser with many tabs used 80% memory]
- **Outcome:** [Resolved / Improved / Escalated]
- **Notes:** Did not end unfamiliar system processes or install updates without authorization.

Log 2: Network connectivity

- **Date / time:** [date]
- **User report (simulated):** "Internet is not working."
- **Connection type:** [Wi-Fi / Ethernet]

Command	Purpose	Result you observed
`ipconfig`	Check IP address, subnet mask, default gateway	[paste summary]
`ping 127.0.0.1`	Test local TCP/IP stack	[replies / failed]
`ping <default gateway>`	Test reachability of the local router	[replies / failed]
`nslookup example.com`	Test DNS name resolution	[resolved / failed]

- **Interpretation:** [e.g., Gateway reachable, DNS lookup failed → investigate DNS settings]
- **Outcome:** [Resolved / Escalated to network team with the above evidence]
- **Reminder:** A failed ping alone does not prove a host is down, since some networks block ICMP.

Log 3: Application not opening (Outlook / Teams)

- **Date / time:** [date]
- **Application:** [Outlook / Teams]
- **Exact error message:** [copy the text exactly]
- **Checks performed:**
  1. Confirmed whether the app is responding or frozen.
  2. Verified internet connectivity (see Log 2).
  3. Closed and reopened the application.
  4. Checked for known updates or service outages where accessible.
- **Scope:** [Only this app / whole system]
- **Outcome:** [Resolved / Escalated]
- **Notes:** No reinstall or system setting change was made before basic checks were complete.

Log 4: Sign-in / locked account (simulated)

- **Date / time:** [date]
- **User report:** "I can't sign in" / "My account is locked."
- **Checks performed:**
  1. Confirmed exact error message and which service is affected.

2. Determined whether the problem is the device, the application, or the user account.
3. Explained the authorized password-reset / account-recovery procedure.
4. (Optional, lab only) Practiced unlocking a test user in an Active Directory lab.

- **Outcome:** [Resolved in lab / Escalated to identity team]
- **Notes:** Never reset real employee passwords or change directory permissions without authorization. Identity should be verified before any reset.

Part 2: Knowledge Base Articles

KB-001: Basic Windows Network Connectivity Checks

- **Symptoms:** User cannot open websites or reach network resources.
- **Initial checks:** Confirm Wi-Fi or Ethernet is connected, airplane mode is off, and the cable or Wi-Fi network is correct.
- **Diagnostic commands:** `ipconfig`, `ping 127.0.0.1`, `ping <default gateway>`, `nslookup example.com`.
- **Interpretation:**
  - Missing or unexpected IP address or gateway suggests a configuration or DHCP issue.
  - Gateway unreachable suggests a local connection or network problem.
  - Ping to an IP works but nslookup fails suggests a DNS problem.
- **Resolution:** [Document the verified fix, e.g., reconnect to Wi-Fi, renew the connection.]
- **Escalation:** Attach the output of `ipconfig`, the ping and nslookup results, connection type, time of issue, and number of users affected.

KB-002: Computer Running Slowly

- **Symptoms:** Laptop is slow, applications freeze or take long to open.
- **Initial checks:** Open Task Manager (Ctrl + Shift + Esc) and review CPU, memory and disk usage.
- **Steps:**
  1. Identify the application using unusually high resources.
  2. Ask the user to save work, then close or restart that application.
  3. Check Windows Update for pending updates; install only if authorized.
  4. Ask when the slowness started and whether anything changed recently.
  5. Re-test and confirm with the user.

- **Do not:** End unfamiliar system processes or install software without approval.
- **Escalation:** Disk at 100% persistently, suspected hardware fault, or suspected malware. Include screenshots of Task Manager.

KB-003: Outlook or Teams Not Opening

- **Symptoms:** Application will not start, hangs, or shows an error.
- **Steps:**
  1. Record the exact error message and application version if visible.
  2. Confirm internet connectivity.
  3. Close the app fully (check Task Manager) and reopen.
  4. Check whether other apps are affected, to decide if it is app-specific or system-wide.
  5. Check for known service outages or updates.
- **Resolution:** [Document what worked.]
- **Escalation:** Error persists after restart, multiple users affected, or the error indicates a profile or licensing problem.

Part 3: Priority Guide (for the Ticket Register)

Priority	When to use	Example
High	Critical business service unavailable or many users affected	Business application down for several employees
Medium	One user or small group significantly affected, workaround may exist	One laptop cannot connect to Wi-Fi
Low	Minor issue, little business impact	Cosmetic issue or how-to question

In real organizations, follow the company's impact-and-urgency matrix.

Part 4: Project Report

1. Objective

To simulate the daily work of an L1 service desk technician: receive an issue, create a ticket, troubleshoot, resolve or escalate, and document the outcome.

2. Environment and tools

- Operating system: [Windows 10/11, physical or VM]
- Tools: Task Manager, Command Prompt (`ipconfig` , `ping` , `nslookup`), Microsoft Excel

3. Scenarios tested

#	Scenario	Result
1	Slow computer / unresponsive app	[result]
2	Network connectivity and DNS	[result]
3	Outlook / Teams not opening	[result]
4	Sign-in / locked account (simulated)	[result]
5	Ticket creation and prioritization	10 sample tickets created
6	Knowledge base documentation	3 articles written

4. Key findings

[Write 3–5 sentences about what you discovered, e.g., most issues were resolved by structured checks before any reinstall or change was needed.]

5. Limitations

- Work was done in a lab or simulated environment, not on production systems.
- No access to a real ticketing platform (e.g., ServiceNow) or company Active Directory unless stated otherwise.
- Sample tickets are illustrative, not real incidents.

6. Lessons learned

- Collect exact error messages and evidence before troubleshooting.
- Check scope first: one app, one device, or many users.
- Do not make risky changes without authorization.
- Clear ticket notes make escalation faster.
- Documenting recurring fixes saves the whole team time.

7. Next steps

Windows 10/11 troubleshooting, Microsoft 365 / Outlook / Teams support, networking basics (TCP/IP, DNS, DHCP, VPN, Wi-Fi), incident management and escalation, then ServiceNow and Active Directory fundamentals.

Part 5: Resume and Interview Notes

Resume entry (use only after you have done the work):

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- Practiced diagnosing Windows OS and application issues using systematic troubleshooting techniques.
- Used `ipconfig`, `ping` and `nslookup` to investigate network connectivity and DNS issues.
- Created and prioritized sample incident tickets in Excel, documenting symptoms, steps, status and resolution or escalation notes.
- Prepared knowledge base documentation for recurring technical issues.

Interview summary: "I completed a practical lab to understand L1 service desk work. I diagnosed Windows and application issues, checked connectivity with Command Prompt tools, and documented findings. I also practiced ticket categorization, prioritization and escalation in Excel, and wrote knowledge base articles. It taught me the value of systematic troubleshooting and clear documentation." Be clear this was a lab project.

Part 6: Five-Day Plan

Day	Focus
1	Windows troubleshooting (Task Manager, responsiveness, updates)
2	Network troubleshooting (ipconfig, ping, nslookup)
3	Application and account scenarios
4	Ticket register in Excel (8–10 tickets, priorities)
5	Knowledge base articles and project report